

1.4 a. Energy: $\frac{1}{8} \#$
Power: Unchanged $\#$

3135/2009 施丞蔚

b. Energy: $(\frac{1}{8})^2 \div 0.0156 \#$
Power: $(\frac{1}{8})^2(\frac{1}{8}) \div 0.00195 \#$

c. Energy: $(50\%)^2 = 0.25 \#$
Power: $(50\%)^2(\frac{1}{8}) = 0.03125 \#$

d. Energy: $(25\%)(25\%) + (25\% \cdot 20\%)(75\%)$
 $= 0.0625 + 0.0375 = 0.1 \#$ (全部的0.1)
 $(0.5W)(25\%) + (0.5W \cdot 20\%)(75\%) = 0.2 (W) \#$

1.5 a. Amdahl's Law: $ExTime_{new} = ExTime_{old} \times \left[(1 - \text{Fraction}_{enhanced}) + \frac{\text{Fraction}_{enhanced}}{\text{Speedup}_{enhanced}} \right]$
 $Speedup_{overall} = \frac{ExTime_{old}}{ExTime_{new}} = \frac{1}{(1 - \text{Fraction}_{enhanced}) + \frac{\text{Fraction}_{enhanced}}{\text{Speedup}_{enhanced}}}$
 $= \frac{1}{(1 - 0.8) + \frac{0.8}{4}} = \frac{1}{0.2 + 0.2} = 2.5 \#$

F: 2.5倍 (∵ 為達 2.5 倍速)

V: 2.5 倍 (∵ 通常假設 F 及 V 呈線性關係)

b. Energy: $[4 \times (\frac{1}{2.5})^2] = 0.64 \#$
Power: $[4 \times (\frac{1}{2.5})^2 \times (\frac{1}{2.5})] = 0.256 \#$

c. $\frac{2 \text{ core} + 2 \text{ ASIC}}{4 \text{ core}} = \frac{2 + 0.2 \times 2}{4} = 0.6 \#$

1.9 a. 60% $\#$

b. $(60\%)(20\%) + (40\%)(90\%) = 0.48$

$1 - 0.48 = 0.52 \#$

c. $P \propto V^2 f$, $(1 - 20\%)^2(1 - 40\%) = 0.384$

$1 - 0.384 = 0.616 \#$

d. $(30\%)(20\%) + (30\%) \cdot 0 + (40\%)(90\%) = 0.42$

$1 - 0.42 = 0.58 \#$

1.10 a. FIT: 每 10^9 小時失敗數 = 100

$\Rightarrow MTTF = \frac{10^9}{100} = 10^7 \text{ (hour)} \#$

b. Availability = $\frac{MTTF}{MTTF + MTTR} = \frac{10^7}{10^7 + 24} \div 1 \#$

1.11 a. $\frac{35}{10000} (1000 \times \frac{1}{3}) \div 11.67 \text{ (days)} \#$

1.12 b.
$$Speedup_{overall} = \frac{ExTime_{old}}{ExTime_{new}} = \frac{1}{(1 - \text{Fraction enhanced}) + \frac{\text{Fraction enhanced}}{Speedup_{enhanced}}}$$

 $\Rightarrow 2 = \frac{1}{(1-x) + \frac{x}{20}} \Rightarrow 20 - 20x + x = 10 \Rightarrow x = \frac{10}{19} \div 52.63\% \#$

c.
$$P' = \frac{\text{新加密執行時間}}{\text{新總執行時間}} = \frac{P / Speedup}{(1-P) + P / Speedup} = \frac{0.526/20}{(1-0.526) + 0.526/20} = 5.3\% \#$$

1.14 a.
$$S = \frac{1}{(1-20\%) + 20\%/20} = 1.111 \#$$

b.
$$S = \frac{1}{(1-P_{FP} - P_{cache}) + P_{FP}/Speedup_{FP} + P_{cache} \times Slowdown}$$

$$= \frac{1}{(1-20\%-10\%) + 20\%/2 + 10\% \times \frac{2}{3}}$$

$$= \frac{1}{0.7 + 0.1 + 0.15} = 1.05 \#$$

c.
$$P'_{FP} = \frac{P_{FP}/Speedup_{FP}}{(1-P_{FP} - P_{cache}) + P_{FP}/Speedup_{FP} + P_{cache} \times Slowdown_{cache}} = \frac{20\%/2}{(1-20\%-10\%) + 20\%/2 + 10\% \times \frac{2}{3}} = \frac{0.1}{0.95} \div 10.5\%$$

$$P'_{cache} = \frac{P_{cache}/Slowdown_{cache}}{(1-P_{FP} - P_{cache}) + P_{FP}/Speedup_{FP} + P_{cache} \times Slowdown_{cache}} = \frac{10\% \times \frac{2}{3}}{(1-20\%-10\%) + 20\%/2 + 10\% \times \frac{2}{3}} = \frac{0.15}{0.95} \div 15.8\%$$

1.15 a.
$$Speedup_A = \frac{1}{(1-50\%) + 50\%/22} \div 1.913 \#$$

b.
$$Speedup_D = \frac{1}{(1-90\%) + 90\%/22} \div 7.096 \#$$

c. $22 \times 41\% \div 9 \text{ (約9核)}$

$$Speedup_{A_{(9core)}} = \frac{1}{(1-50\%) + 50\%/9} = 1.8 \#$$

$$Speedup_{overall} = \frac{1}{(1-41\%) + 41\%/1.8} \div 1.222 \#$$

d.
$$Speedup_A = \frac{1}{(1-50\%) + 50\%/(22 \times 41\%)} \div 1.8 \#$$

$$Speedup_B = \frac{1}{(1-80\%) + 80\%/(22 \times 27\%)} \div 2.988 \#$$

$$Speedup_C = \frac{1}{(1-60\%) + 60\%/(22 \times 18\%)} \div 1.813 \#$$

$$Speedup_D = \frac{1}{(1-90\%) + 90\%/(22 \times 14\%)} \div 2.55 \#$$

$$Speedup_{overall} = \frac{1}{\frac{41\%}{1.8} + \frac{27\%}{2.988} + \frac{18\%}{1.813} + \frac{14\%}{2.55}} = 2.117 \#$$

e.
$$P'_A = \frac{\frac{41\%}{1.8}}{\frac{41\%}{1.8} + \frac{27\%}{2.988} + \frac{18\%}{1.813} + \frac{14\%}{2.55}} \div 48\%$$

$$P'_B = \frac{\frac{27\%}{2.988}}{\frac{41\%}{1.8} + \frac{27\%}{2.988} + \frac{18\%}{1.813} + \frac{14\%}{2.55}} \div 19\%$$

$$P'_C = \frac{\frac{18\%}{1.813}}{\frac{41\%}{1.8} + \frac{27\%}{2.988} + \frac{18\%}{1.813} + \frac{14\%}{2.55}} \div 21\%$$

$$P'_D = \frac{\frac{14\%}{2.55}}{\frac{41\%}{1.8} + \frac{27\%}{2.988} + \frac{18\%}{1.813} + \frac{14\%}{2.55}} \div 12\%$$

1.16 a. $Speedup_N = \frac{1}{(1-80\%) + 80\%/N} = \frac{1}{0.2 + 0.8/N} \#$

b. $Speedup = \frac{1}{(1-P) + \text{通訊開銷} + P/N} = \frac{1}{(1-80\%) + 0.05\% \times 8 + 80\%/8} \div 2.94 \#$

c. 8個 processors $\Rightarrow 2^3$ (加倍3次), $\log_2 8 = 3$

$Speedup = \frac{1}{(1-80\%) + 0.05\% \times 3 + 80\%/8} \div 3.17 \#$

d. $Speedup = \frac{1}{(1-80\%) + 0.05\% \times \log_2 N + 80\%/N} = \frac{1}{0.2 + 0.005 \log_2 N + 0.8/N} \#$

e. $\frac{d}{dN} \left(\frac{1}{(1-P) + 0.005 \log_2 N + P/N} \right) = 0 \#$